Gorka Fraga González

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Experience

Scientific Staff | Data Steward — 2024 - Present
Center for Reproducible Science and Research Synthesis,
Epidemiology, Biostatistics and Prevention Institute, University of Zurich
- Lead training workshops on reproducibility in practice (Git, dynamic reporting)
- Mentor and support researchers in various aspects of data management
- Lead project COCOPREND to develop a data code of conduct
- Data steward at Linguistic Research Infrastructure (LiRI)

Scientific Staff — Aug 2023 - 2024
Center for Reproducible Science and Research Synthesis,
Epidemiology, Biostatistics and Prevention Institute, University of Zurich
Co-lead project AFFORD: A Framework for Avoiding the Open Research Data Dump
- Developed guides, workflows and tools for Open Research Data
- Lead data curation support in a preclinical neuroimaging Sinergia project

Postdoctoral Researcher — May 2022 - Aug 2023
Neurolinguistics Division, Psychology Institute, University of Zurich
Project on brain states associated with speech-in-noise comprehension
- Designed and implemented EEG and online experiments
- Set up lab's Git repository, website (markdown), guidelines (jupyter-book)

Postdoctoral Researcher - Project Leader — Mar 2020 -Mar 2022
Department of Child and Adolescent Psychiatry, University of Zurich
Project 'Grapholemo: grapheme-phoneme learning modeling'
- Applied reinforcement learning and diffusion decision models (R-Stan)
- Managed study database (Redcap)
- Support doctoral students: code, statistics, documentation, publishing

Postdoctoral Researcher — May 2018 - Mar 2020
Department of Child and Adolescent Psychiatry, University of Zurich
Longitudinal assessment of a digital app to support struggling readers
- Created Matlab scripts for signal processing and data analysis (fMRI, EEG data)
- Analyzed neural, digital game performance and cognitive tests (R, SPSS)

Postdoctoral Researcher — Jan 2016 - Mar 2018
Developmental Psychology, University of Amsterdam.
Using network analysis to examine brain function in dyslexia
- Collected and analyzed electrophysiological and psychometric data

PhD — Sept 2011 - Dec 2015
Developmental Psychology, University of Amsterdam.
Examining neural and behavioral success factor of a dyslexia intervention
- Coordinated a longitudinal neuroscientific study in 3rd grade children
- Psychometric and EEG data preprocessing and analysis (Matlab,SPSS)
- Clinical trial design, implementation, analysis and reporting

Education

- CAS (12 ECTS) on Data Stewardship, University of Lausanne, Switzerland 2024 - 2025
- PhD in Developmental Psychology. University of Amsterdam, The Netherlands 2011 - 2015
- MSc in Psychology (cum laude). University of Groningen, The Netherlands 2010 - 2011
- BSc in Psychology. University of Deusto, Bilbao, Spain 2003 - 2008

Leadership and national strategy engagement

- Lead of the Working Group on Open Research Data from the Swiss Reproducibility Network
- Co-lead of local node (UZH) in the Swiss Reproducibility Network
- Co-lead of "Strategic Orientation 4: internationalization" from the "Blueprint Level B: operationalizing the strategic orientations for the health and life sciences cluster in Switzerland". Report supported by the National ORD Strategy Council
- Contribution to "Action Line B5.1 establish best ORD practices" from the National ORD action plan mandated by swissuniversities. Lead landscape analysis in the final report
- Member of Swiss Research Data Support network and Data Stewards Network of UZH

Technical Skills

- Software:
  - Advanced programming: R, Python, Matlab
  - Quarto, R markdown, Jupyter Notebooks, LaTeX, Git
  - MS Office, Redcap, SPSS, neuroimaging (MNE, SPM, EEGlab, FSL, BrainVision)
- Statistics and machine learning: regression, mixed models, reinforcement learning, drift diffusion
- Experimental design: cross-sectional, longitudinal training studies, clinical trials
- Neuroscience methods: electroencephalography, electrocardiography, magnetic resonance imaging

Further experience

- 20+ peer-reviewed publications, including data management guidelines and workflows
- Organized multiple outreach activities (international symposiums, webinars, workshops)
- 3 Awarded postdoctoral personal grants
- Mentoring at Bachelor, Master and Doctoral level
- Best science poster, SIG22 Neuroscience and Education conference, London (2018)
- Oral communication in international conferences (USA and Europe)

Professional training

- Fundamentals of scientific metadata, University of Basel 2024
- Advanced neural time series analysis with Matlab
- R programming for data science
- Human neuroanatomy

Languages

- Spanish (native)
- English (full professional competence)
- German (intermediate)
- Dutch (basic)

Other Skills

- Critical thinker and detail-oriented
- Ability to mediate between technical staff, researchers and other stakeholders
- Fast learner of new software and programming, enthusiastic about data visualization and design
- I enjoy working in international and multidisciplinary teams

Scientific Contributions

Open Research Data | Data Management

2025 **Fraga-González, G**., van de Wiel, H., Garassino, F., Kuo, W., de Zélicourt, D., Kurtcuoglu, V., ... & Furrer, E. (2025). Affording reusable data: recommendations for researchers from a data-intensive project. *Scientific Data*, 12(1), 258. https://doi.org/10.1038/s41597-025-04565-0

2024 **Fraga González, G**., Garassino, F., Wiel, H. v., Kuo, W., de Zélicourt, D. d. J., Kurtcuoglu, V., … Held, L. (2024, December 12). Where are my data? The AFFORD workflow to create your own data index with Git, R and Quarto. *MetArXiv*. https://doi.org/10.31222/osf.io/64fch

Wiel, H. v., Garassino, F., Li, Z., **Fraga González, G.**, Furrer, E., & Held, L. (2024, December 23). Stakeholder Engagement for Sustainable Open Research Data Support Services: Insights from Interviews and Surveys in Switzerland. *MetaArXiv*. https://doi.org/10.31222/osf.io/3d5we

Neuroscience

2025 **Fraga-González, G**., Haller, P., Willinger, D., Gehrig, V., Frei, N., & Brem, S. (2025). Functional brain activations correlated with association strength and prediction error during novel symbol–speech sound learning. *Imaging Neuroscience*, 3. https://doi.org/10.1162/imag_a_00439

2024 Lutz, C. G., Coraj, S., **Fraga-González, G.**, & Brem, S. (2024). The odd one out–Orthographic oddball processing in children with poor versus typical reading skills in a fast periodic visual stimulation EEG paradigm. *Cortex*, 172, 185-203. https://doi.org/10.1016/j.cortex.2023.12.010

2022 **Fraga-González, G**., Jednoróg, K., & Brem, S. (2022). Longitudinal neural observation studies of dyslexia. In M.A. Skeide (Ed), *The Cambridge Handbook of Dyslexia and Dyscalculia*. Cambridge University Press. https://doi.org/10.1017/9781108973595

Fraga-González, G., Di Pietro, S. V., Pleisch, G., Walitza, S., Brandeis, D., Karipidis, I. I., & Brem, S. (2022). Visual Occipito-Temporal N1 Sensitivity to Digits Across Elementary School. *Frontiers in Human Neuroscience*, 44. https://doi.org/10.3389/fnhum.2022.887413

Zhang, M., Riecke, L., **Fraga-González, G.**, & Bonte, M. (2022). Altered brain network topology during speech tracking in developmental dyslexia. *NeuroImage*, 254, 119142. https://doi.org/10.1016/j.neuroimage.2022.119142

2021 **Fraga González G,** Smit DJA, Van der Molen MJW, Tijms J, Stam CJ, de Geus EJC and Van der Molen MW. (2021). Graph Analysis of EEG Functional Connectivity Networks During a Letter-Speech Sound Binding Task in Adult Dyslexics. *Frontiers in Psychology*, 12. https://doi.org/10.3389/fpsyg.2021.767839

Fraga-González, G., Pleisch, G., Di Pietro, S. V., Neuenschwander, J., Walitza, S., Brandeis, D., ... & Brem, S. (2021). The rise and fall of rapid occipito-temporal sensitivity to letters: Transient specialization through elementary school. *Developmental Cognitive Neuroscience*, 49, 100958. https://doi.org/10.1016/j.dcn.2021.100958

Duerler P, Brem S**, Fraga-González G**, Neef T, Allen M, Zeidman P, Stämpfli P, Vollenweider FX, Preller KH. (2021). Psilocybin Induces Aberrant Prediction Error Processing of Tactile Mismatch Responses-A Simultaneous EEG-FMRI Study. *Cerebral Cortex*, 31(12), 5556-5565. https://doi.org/10.1093/cercor/bhab202

Chyl, K., Fraga-González*, G., Brem, S., & Jednoróg, K. (2021). Brain dynamics of (a) typical reading development—a review of longitudinal studies. *npj Science of Learning*, 6(1), 1-9. https://doi.org/10.1038/s41539-020-00081-5 *First authorship shared

2020 Mehringer, H., **Fraga-González***, G., Pleisch, G., Röthlisberger, M., Aepli, F., Keller, V., ... & Brem, S. (2020). (Swiss) GraphoLearn: an app-based tool to support beginning readers. *Research and Practice in Technology Enhanced Learning*, 15(1), 1-21. https://doi.org/10.1186/s41039-020-0125-0 *First authorship shared

Liu, Y., van den Wildenberg, W. P., **Fraga-González, G.**, Rigoni, D., Brass, M., Wiers, R. W., & Ridderinkhof, K. R. (2020). "Free won't" after a beer or two: chronic and acute effects of alcohol on neural and behavioral indices of intentional inhibition. *BMC Psychology*, 8(1), 1-20. https://doi.org/10.1186/s40359-019-0367-z

Wang, F., Karipidis, I. I., Pleisch, G., **Fraga-González, G.**, & Brem, S. (2020). Development of print-speech integration in the brain of beginning readers with varying reading skills. *Frontiers in Human Neuroscience*, 14, 289. https://doi.org/10.3389/fnhum.2020.00289

Tijms, J., **Fraga-González, G**., Karipidis, I. I., & Brem, S. (2020). Editorial: The Role of Letter-Speech Sound Integration in Normal and Abnormal Reading Development. *Frontiers in Psychology*, 11, 1441. https://doi.org/10.3389/fpsyg.2020.01441

2019 **Fraga González, G**., Smit, D. J. A., van der Molen, M. J. W., Tijms, J., de Geus, E. J. C., & van der Molen, M. W. (2019). Probability learning and feedback processing in dyslexia: A performance and heart rate analysis. *Psychophysiology*, 56(12), e13457https://doi.org/10.1111/psyp.13460

Goikoetxea, E., Van Bon, W., **Fraga González, G**., & Martínez, N. (2019). Psychometric properties of a silent word reading test (LEO-1-min). *Revista Electrónica de Investigación Psicoeducativa y Psicopedagógica*, 17(48). https://doi.org/10.25115/ejrep.v17i48.2181

2018 **Fraga González, G.**, Karipidis, I., & Tijms, J. (2018). Dyslexia as a neurodevelopmental disorder and what makes it different from a chess disorder. *Brain Sciences*, 8(10), 189. https://doi.org/10.3390/brainsci8100189

Fraga González, G., Smit, D. J., Van Der Molen, M. J., Tijms, J., Stam, C. J., De Geus, E., & Van Der Molen, M. W. (2018). EEG resting state functional connectivity in adult dyslexics using phase lag index and graph analysis. *Frontiers in Human Neuroscience*, 12, 341. https://doi.org/10.3389/fnhum.2018.00341

Žarić, G., Timmers, I., Gerretsen, P., **Fraga González, G**., Tijms, J., van der Molen, M. W., ... & Bonte, M. (2018). Atypical white matter connectivity in dyslexic readers of a fairly transparent orthography. *Frontiers in Psychology*, 9, 1147. https://doi.org/10.3389/fpsyg.2018.01147

2017 **Fraga González, G.***, Žarić, G., Tijms, J., Bonte, M., & van der Molen, M. W. (2017). Contributions of letter-speech sound learning and visual print tuning to reading improvement: evidence from brain potential and dyslexia training studies. *Brain Sciences*, 7(1), 10. https://doi.org/10.3390/brainsci7010010

Žarić, G., Correia, J. M., **Fraga González, G.,** Tijms, J., van der Molen, M. W., Blomert, L., & Bonte, M. (2017). Altered patterns of directed connectivity within the reading network of dyslexic children and their relation to reading dysfluency. *Developmental Cognitive Neuroscience*, 23, 1-13. https://doi.org/10.1016/j.dcn.2016.11.003

2016 Fraga González G, van der Molen MW, Žarić G, Bonte M, Tijms J, Blomert L, et al. (2016). Graph analysis of EEG resting state functional networks in dyslexic readers. [Corrigendum published in Clinical Neurophysiology. 129(1) (2018) 339-340] *Clinical Neurophysiology*, 127, 3165–3175. https://doi.org/10.1016/j.clinph.2016.06.023

Fraga González, G., Žarić, G., Tijms, J., Bonte, M., Blomert, L., Leppänen, P.H.T., & van der Molen, M. W. (2016). Responsivity to dyslexia training indexed by the N170 amplitude of the brain potential elicited by word reading. *Brain and Cognition*, 106, 42-54. https://doi.org/10.1016/j.bandc.2016.05.001

2015 Fraga González, G., Žarić G, Tijms J, Bonte M, Blomert L, van der Molen MW (2015). A randomized controlled trial on the beneficial effects of training letter-speech sound integration on reading fluency in children with dyslexia. *PLoS ONE*, 10(12), e0143914. https://doi.org/10.1371/journal.pone.0143914

Žarić, G., **Fraga González, G.,** Tijms, J., van der Molen, M.W., Blomert, L., Bonte, M. (2015). Crossmodal deficit in dyslexic children: practice affects the neural timing of letter-speech sound integration. *Frontiers in Human Neuroscience*, 9, 369. https://doi.org/10.3389/fnhum.2015.00369

2014 Fraga González, G., Žarić, G., Tijms, J., Blomert, L., Bonte, M., & van der Molen, M. W. (2014). Brain-potential analysis of visual word recognition in dyslexics and typically reading children. *Frontiers in Human Neuroscience*, 8, 474. https://doi.org/10.3389/fnhum.2014.00474

Žarić, G.**, Fraga González, G.**, Tijms, J., van der Molen, M. W., Blomert, L., & Bonte, M. (2014). Reduced neural integration of letters and speech sounds in dyslexic children scales with individual differences in reading fluency. *PLoS ONE*, 9(10), e110337. https://doi.org/10.1371/journal.pone.0110337